Title

Sim2Terrain AI

Team Name : IgniteX

Project Name:

Sim2TerrainAI: Synthetic Data Driven Off-Road Segmentation for Autonomous Navigation

Brief Tagline:

An AI-driven system that uses synthetic simulation data to train semantic segmentation models for accurate terrain understanding and safe autonomous navigation in complex off-road environments.

METHODOLOGY

The proposed system follows a structured pipeline to train and evaluate a semantic segmentation model for off-road terrain understanding. The process begins with generating synthetic terrain data using the Falcon Digital Twin Simulation platform. This platform enables the creation of labeled desert terrain images representing different environmental conditions such as sand, rocks, vegetation, and obstacles.

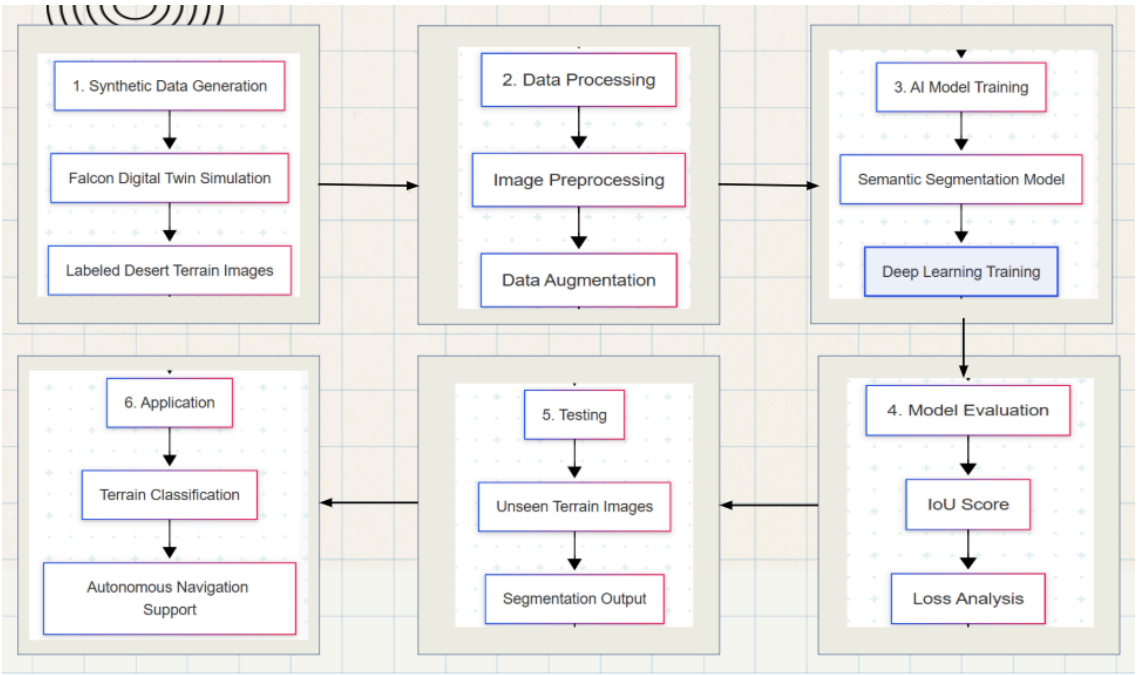
The dataset was generated using Duality Al's Falcon simulation platform, which creates high-fidelity geospatial-based digital twin environments of desert terrain. The dataset consists of RGB color images paired with pixel-level segmentation masks, where each pixel is labeled with one of 10 terrain classes:

Class ID	Class Name	Description
100	Trees	Desert trees and cactus
200	Lush Bushes	Green/leafy dense shrubs
300	Dry Grass	Arid, yellowed grass patches
500	Dry Bushes	Sparse, brown desert shrubs
550	Ground Clutter	Mixed small debris on ground
600	Flowers	Desert flowers and low plants
700	Logs	Fallen logs and dead wood
800	Rocks	Rocks and stone formations
7100	Landscape	General ground / sand / soil
10000	Sky	Sky and atmospheric regions

The training dataset contains 2,857 images, the validation split contains 317 images (both from the

same desert environment), and the test set contains 1,002 images from a novel, unseen desert environment — specifically designed to evaluate cross-environment generalization.

Workflow Diagram:



The workflow illustrates the complete pipeline starting from synthetic data generation to model training and evaluation. The Falcon simulation platform generates labeled terrain data which is then preprocessed and used to train the DeepLabV3 segmentation model. The trained model is evaluated using IoU scores and loss analysis before being tested on unseen environments.

Fine-Tuning and Optimization :

To improve the segmentation performance, several fine-tuning strategies were applied during training. Hyperparameters such as learning rate and batch size were optimized to achieve better convergence. Transfer learning was used by initializing the ResNet-101 backbone with pre-trained weights, allowing the model to learn terrain features more effectively.

Additionally, data augmentation techniques helped increase dataset diversity and improved the model's ability to generalize across different terrain conditions. These optimizations resulted in improved segmentation accuracy and stable training performance.

RESULT & PERFORMANCE MATRICS

The performance of the proposed terrain segmentation model was evaluated using standard semantic segmentation metrics. The primary evaluation metric used in this project is Intersection over Union (IoU), which measures the overlap between the predicted segmentation and the ground truth labels.

A higher IoU score indicates better segmentation accuracy and improved terrain classification capability. The model was trained using the synthetic dataset generated from the Falcon Digital Twin simulation platform and evaluated across different terrain classes such as sand, rocks, vegetation, and obstacles.

In addition to IoU, a confusion matrix was used to analyze the classification performance across terrain categories. The confusion matrix helps identify correct predictions as well as misclassifications between terrain classes.

The evaluation results demonstrate that the model is capable of accurately identifying different terrain regions while maintaining stable training performance.

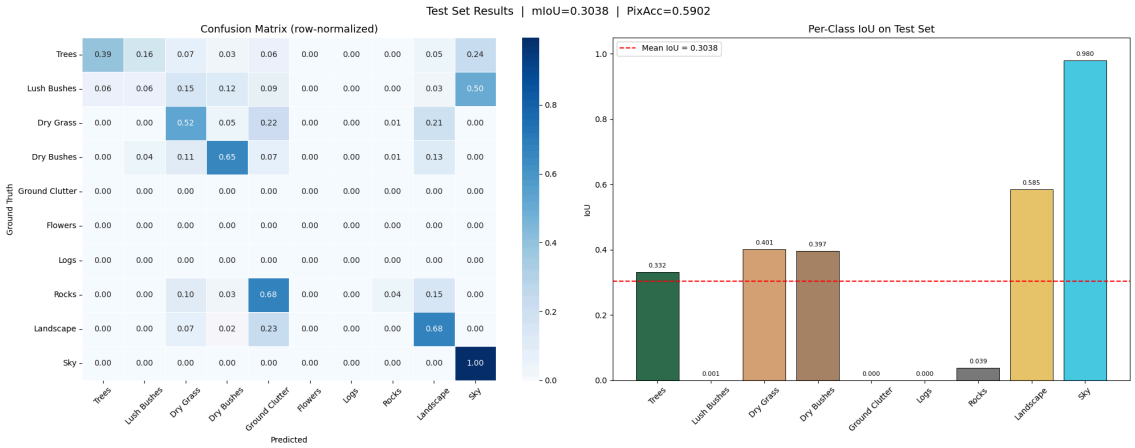


Fig : Confusion Matrix

Accuracy Comparison and Model Performance :

To further analyze model performance, accuracy trends were evaluated during the training process. The training loss graph shows how the error decreases over epochs, indicating that the model gradually learns meaningful terrain features.

An accuracy comparison analysis was also performed to observe the segmentation performance across different terrain classes. The results show that the model performs well in identifying dominant terrain regions such as sand and vegetation while maintaining reasonable accuracy in more complex terrain categories.

These results confirm that the trained semantic segmentation model can generalize well when applied to unseen terrain environments.

```
    Landscape        : 0.5407
    Sky              : 0.9752

Ep 011/50 | TrLoss 1.1267 | VaLoss 1.0667 | TrAcc 0.7621 | VaAcc 0.8068 | mIoU 0.4903 | LR 4.8e-05 | 206s  ⭐ NEW BEST
Ep 012/50 | TrLoss 1.1208 | VaLoss 1.0634 | TrAcc 0.7656 | VaAcc 0.8105 | mIoU 0.4969 | LR 4.7e-05 | 207s  ⭐ NEW BEST
Ep 013/50 | TrLoss 1.1150 | VaLoss 1.0603 | TrAcc 0.7678 | VaAcc 0.8120 | mIoU 0.5017 | LR 4.6e-05 | 206s  ⭐ NEW BEST
Ep 014/50 | TrLoss 1.1133 | VaLoss 1.0600 | TrAcc 0.7690 | VaAcc 0.8112 | mIoU 0.5042 | LR 4.5e-05 | 206s  ⭐ NEW BEST
Ep 015/50 | TrLoss 1.1055 | VaLoss 1.0547 | TrAcc 0.7730 | VaAcc 0.8126 | mIoU 0.5115 | LR 4.4e-05 | 207s  ⭐ NEW BEST
  Per-class IoU:
    Trees            : 0.7630
    Lush Bushes      : 0.5881
    Dry Grass        : 0.6227
    Dry Bushes       : 0.3887
    Ground Clutter   : 0.2973
    Flowers          : 0.4910
    Logs             : 0.0935
    Rocks            : 0.3387
    Landscape        : 0.5553
    Sky              : 0.9766

Ep 016/50 | TrLoss 1.1000 | VaLoss 1.0517 | TrAcc 0.7756 | VaAcc 0.8067 | mIoU 0.5131 | LR 4.3e-05 | 206s  ⭐ NEW BEST
Ep 017/50 | TrLoss 1.0986 | VaLoss 1.0529 | TrAcc 0.7773 | VaAcc 0.8030 | mIoU 0.5163 | LR 4.2e-05 | 206s  ⭐ NEW BEST
Ep 018/50 | TrLoss 1.0924 | VaLoss 1.0464 | TrAcc 0.7794 | VaAcc 0.8133 | mIoU 0.5281 | LR 4.1e-05 | 207s  ⭐ NEW BEST
Ep 019/50 | TrLoss 1.0899 | VaLoss 1.0461 | TrAcc 0.7801 | VaAcc 0.8049 | mIoU 0.5230 | LR 3.9e-05 | 206s
Ep 020/50 | TrLoss 1.0875 | VaLoss 1.0433 | TrAcc 0.7811 | VaAcc 0.8135 | mIoU 0.5358 | LR 3.8e-05 | 207s  ⭐ NEW BEST
  Per-class IoU:
    Trees            : 0.7712
    Lush Bushes      : 0.5940
    Dry Grass        : 0.6157
    Dry Bushes       : 0.3905
    Ground Clutter   : 0.2975
    Flowers          : 0.5260
    Logs             : 0.2673
    Rocks            : 0.3579
    Landscape        : 0.5616
    Sky              : 0.9761

Ep 021/50 | TrLoss 1.0840 | VaLoss 1.0422 | TrAcc 0.7822 | VaAcc 0.8165 | mIoU 0.5416 | LR 3.6e-05 | 207s  ⭐ NEW BEST
Ep 022/50 | TrLoss 1.0827 | VaLoss 1.0398 | TrAcc 0.7829 | VaAcc 0.8113 | mIoU 0.5385 | LR 3.5e-05 | 205s
Ep 023/50 | TrLoss 1.0790 | VaLoss 1.0380 | TrAcc 0.7860 | VaAcc 0.8132 | mIoU 0.5379 | LR 3.3e-05 | 205s
Ep 024/50 | TrLoss 1.0774 | VaLoss 1.0374 | TrAcc 0.7862 | VaAcc 0.8115 | mIoU 0.5401 | LR 3.1e-05 | 205s
Ep 025/50 | TrLoss 1.0741 | VaLoss 1.0358 | TrAcc 0.7890 | VaAcc 0.8119 | mIoU 0.5398 | LR 3.0e-05 | 205s
  Per-class IoU:
```

CHALLENGES

1. When i was importing the dataset in google colab the Dataset Drive link limit is reached so we were unable to upload using Gdown
2. Training The Neural Networks with huge dataset and and lot of epochs with multiple models for testing the scores we need lot of computing power so we switched to collab gpu but the limit reached in 3hrs by using T4 gpu
3. When we first trained the model we got these scores. When we trained further with more epochs our “Best Model” according to your training process did not

```
Starting evaluation on test data...
Evaluation complete. Calculating metrics...

--- Test Metrics ---
Pixel Accuracy: 0.5582
Mean Precision: 0.4325
Mean Recall:    0.3418
Mean F1 Score:  0.3421
mIoU:           0.2630
--------------------
Per-class metrics:
Class 0: Precision=0.7385, Recall=0.2725, F1=0.3981, IoU=0.2485
Class 1: Precision=0.0000, Recall=0.0021, F1=0.0000, IoU=0.0000
Class 2: Precision=0.7034, Recall=0.2514, F1=0.3704, IoU=0.2273
Class 3: Precision=0.0000, Recall=0.0000, F1=0.0000, IoU=0.0000
Class 4: Precision=0.0000, Recall=0.0000, F1=0.0000, IoU=0.0000
Class 5: Precision=0.0000, Recall=0.0000, F1=0.0000, IoU=0.0000
Class 6: Precision=0.3072, Recall=0.7837, F1=0.4414, IoU=0.2832
Class 7: Precision=0.7342, Recall=0.4257, F1=0.5390, IoU=0.3689
Class 8: Precision=0.9769, Recall=0.9988, F1=0.9877, IoU=0.9757
```

```
Starting evaluation of the best model on test data...
Evaluation complete. Calculating metrics...

--- Test Metrics (Best Model) ---
Pixel Accuracy: 0.5562
Mean Precision: 0.3693
Mean Recall:    0.3090
Mean F1 Score:  0.2742
mIoU:           0.2199
--------------------
Per-class metrics:
Class 0: Precision=0.6150, Recall=0.1687, F1=0.2647, IoU=0.1526
Class 1: Precision=0.0000, Recall=0.2367, F1=0.0001, IoU=0.0000
Class 2: Precision=0.3828, Recall=0.8360, F1=0.5252, IoU=0.3561
Class 3: Precision=0.0000, Recall=0.0000, F1=0.0000, IoU=0.0000
Class 4: Precision=0.0000, Recall=0.0000, F1=0.0000, IoU=0.0000
Class 5: Precision=0.0000, Recall=0.0000, F1=0.0000, IoU=0.0000
Class 6: Precision=0.5321, Recall=0.0323, F1=0.0610, IoU=0.0314
Class 7: Precision=0.8068, Recall=0.5096, F1=0.6246, IoU=0.4541
Class 8: Precision=0.9873, Recall=0.9974, F1=0.9923, IoU=0.9848
```

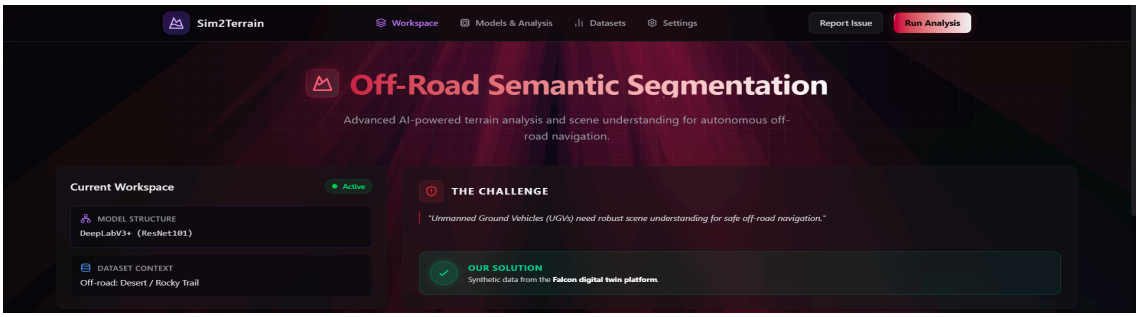
   generalize as well on the test set.
4. So then we change the approach we use pretrain model + fine-tuning it.

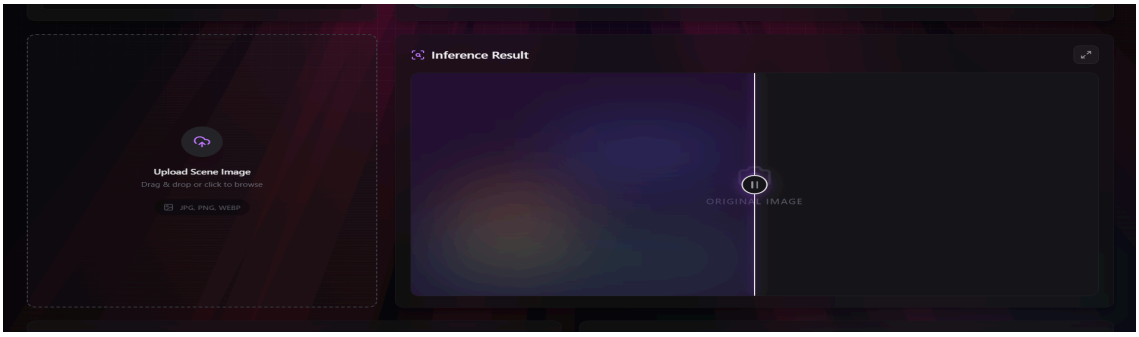
   DeepLabV3+ (ResNet50 pretrained) | Dice+CE Loss | AdamW + Cosine LR

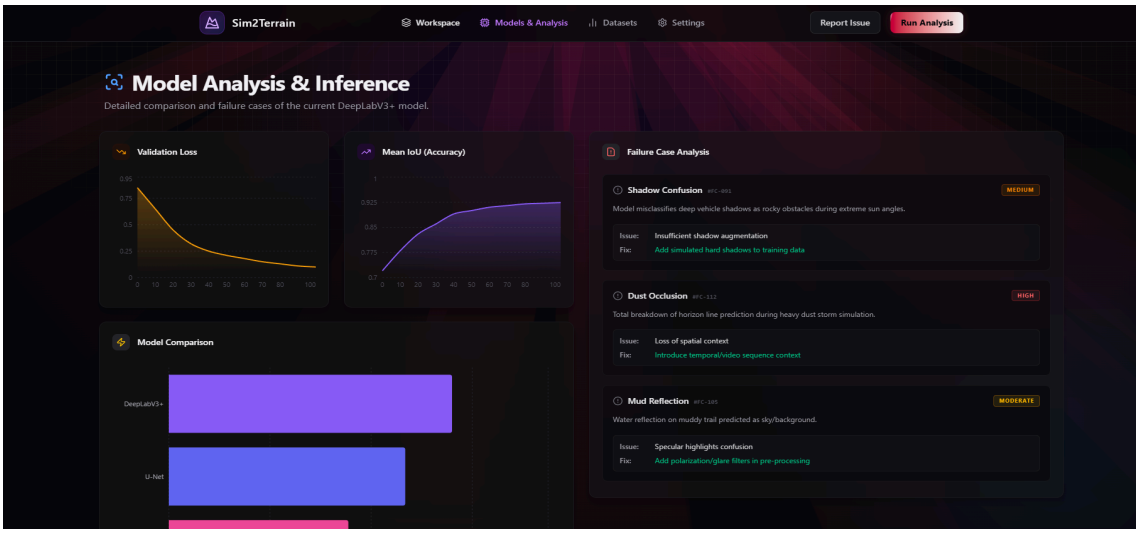
5. The mIOU score started increasing.

Solution

Validation: Ground Truth vs Prediction
Input
Ground Truth
Prediction
Trees
Lush Bushes
Dry Grass
Dry Bushes
Ground Clutter
Flowers
Logs
Rocks
Landscape
Sky


Sim2Terrain
Workspace
Models & Analysis
Datasets
Settings
Report Issue
Run Analysis
Off-Road Semantic Segmentation
Advanced AI-powered terrain analysis and scene understanding for autonomous off-road navigation.
Current Workspace
Active
MODEL STRUCTURE
DeepLabV3+ (ResNet101)
DATASET CONTEXT
Off-road: Desert / Rocky Trail
THE CHALLENGE
"Unmanned Ground Vehicles (UGVs) need robust scene understanding for safe off-road navigation."
OUR SOLUTION
Synthetic data from the Falcon digital twin platform.


Upload Scene Image
Drag & drop or click to browse
JPG, PNG, WEBP
Inference Result
ORIGINAL IMAGE


Sim2Terrain
Workspace
Models & Analysis
Datasets
Settings
Report Issue
Run Analysis
Model Analysis & Inference
Detailed comparison and failure cases of the current DeepLabV3+ model.
Validation Loss
Mean IoU (Accuracy)
Failure Case Analysis
Shadow Confusion
MEDIUM
Model misclassifies deep vehicle shadows as rocky obstacles during extreme sun angles.
Issue: Insufficient shadow augmentation
Fix: Add simulated hard shadows to training data
Dust Occlusion
HIGH
Total breakdown of horizon line prediction during heavy dust storm simulation.
Issue: Loss of spatial context
Fix: Introduce temporal/video sequence context
Mud Reflection
MODERATE
Water reflection on muddy trail predicted as sky/background.
Issue: Specular highlights confusion
Fix: Add polarization/glare filters in pre-processing
Model Comparison
DeepLabV3+
U-Net

Inference Result

FUTURE WORK

Future improvements could include:

- In car built in applications
- Military grade (BSF) in desert terrains like Rajasthan
- Using larger and more diverse datasets
- Applying advanced data augmentation techniques
- Training more advanced architectures such as **DeepLabV3+**
- Improving model performance for rare classes

These improvements could further enhance the robustness and accuracy of the segmentation model.

CONCLUSION

This project demonstrates the effectiveness of deep learning-based semantic segmentation for off-road terrain analysis.

By using the DeepLabV3 architecture and synthetic data generated through digital twin environments, the model was able to identify different terrain elements with good accuracy.

The final model achieved a validation IoU score of approximately , demonstrating promising results for off-road perception tasks.